

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-1977

Sierra Pacific Lumber 345 kV
July 18, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1977. The Interconnection Customer proposes a 129.9 MW battery energy storage facility interconnecting at the Sierra Pacific Lumber 345 kV substation in the Maryland area. The request was received on January 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 563 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 31 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,679,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SANDARIO-FOX RIV 138 kV line
Contingency: SIERRA PACIFIC LUMBER-LAKE CREEK 138 kV Line
Loading: 160.0% (Rating: 171 MVA)
DFAX: 14.1%
Pre-project loading: 149.4%
2. Facility: SIERRA PACIFIC LUMBER-CALIFORNIA AVE 138 kV L
Contingency: LAKE CRK-SIERRA PACIFIC LUMBER 138 kV Line
Loading: 158.8% (Rating: 233 MVA)
DFAX: 14.1%
Pre-project loading: 151.0%
3. Facility: CHANNELVIEW-CALIFORNIA AVE SUBSTATION 138 kV
Contingency: LAKE CRK-SIERRA PACIFIC LUMBER 138 kV
Loading: 152.8% (Rating: 220 MVA)
DFAX: 14.1%
Pre-project loading: 144.5%
4. Facility: GEORGIA-PACIFIC CROSSETT LLC SUBSTATION-CHANNELVIEW 138 kV
Contingency: SIERRA PACIFIC LUMBER-LAKE CREEK 138 kV
Loading: 149.0% (Rating: 206 MVA)
DFAX: 14.1%
Pre-project loading: 140.1%
5. Facility: FOX RIV-GEORGIA-PACIFIC CROSSETT LLC 138 kV L
Contingency: LAKE CRK-SIERRA PACIFIC LUMBER 138 kV
Loading: 148.4% (Rating: 200 MVA)
DFAX: 14.1%
Pre-project loading: 139.3%

6. Facility: FOOTHILL-WAUGH CHAPEL 138 kV
Contingency: RIVA RD-SIERRA PCIFIC LUMBER 345 kV line
Loading: 148.1% (Rating: 190 MVA)
DFAX: 13.4%
Pre-project loading: 138.9%
7. Facility: GLEN BURNIE SUBSTATION-SANDARIO 138 kV
Contingency: LAKE CREEK-SIERRA PACIFIC LUMBER 138 kV
Loading: 143.7% (Rating: 305 MVA)
DFAX: 8.6%
Pre-project loading: 140.1%
8. Facility: FOOTHILL-MCCALL SUBSTTAION 138 kV
Contingency: SIERRA PACIFIC LUMBER SUBSTATION-RIVA ROAD 345 kV
Loading: 136.1% (Rating: 235 MVA)
DFAX: 13.4%
Pre-project loading: 128.6%
9. Facility: HUNT CLUB-SIERRA PACIFIC LUMBER 138 kV L
Contingency: LAKE CRK-SIERRA PACIFIC LUMBER 138 kV
Loading: 135.3% (Rating: 232 MVA)
DFAX: 9.1%
Pre-project loading: 130.2%
10. Facility: HUNT CLUB-SALAMANCAFRANK ST 138 kV
Contingency: LAKE CRK-SIERRA PACIFIC LUMBER 138 kV
Loading: 132.4% (Rating: 225 MVA)
DFAX: 9.1%
Pre-project loading: 127.1%
11. Facility: STILLWATER HYDRO ELECTRIC PROJECT-SANDARIO 138 kV
Contingency: RIVA ROAD-SIERRA PACIFIC LUMBER 345 kV
Loading: 129.2% (Rating: 343 MVA)
DFAX: 22.5%
Pre-project loading: 120.7%
12. Facility: MCCALL SUBSTTAION-PEBBLEDALE SUBSTATION 138 kV
Contingency: RIVA ROAD-SIERRA PACIFIC LUMBER 345 kV
Loading: 128.6% (Rating: 635 MVA)
DFAX: 35.9%
Pre-project loading: 121.2%
13. Facility: ALLIANCE SUBSTATION-RO BLANCO 2702 SUBSTATION 138 kV
Contingency: RIVA RD-SIERRA PACIFIC LUMBER SUBSTATION 345 kV
Loading: 128.2% (Rating: 404 MVA)
DFAX: 30.4%
Pre-project loading: 118.4%
14. Facility: MCCALL SUBSTTAION-STILLWATER HYDRO ELECTRIC PROJECT 138 kV
Contingency: RIVA ROAD-SIERRA PACIFIC LUMBER 345 kV Line
Loading: 127.2% (Rating: 364 MVA)
DFAX: 22.5%
Pre-project loading: 119.1%
15. Facility: BRUCE CROSSING SUBSTATION-RIO BLANCO 2702 138 kV
Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV
Loading: 125.9% (Rating: 433 MVA)
DFAX: 30.4%
Pre-project loading: 116.8%
16. Facility: TABER-PEBBLEDALE 138 kV
Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV

Loading: 125.9% (Rating: 687 MVA)
DFAX: 35.9%
Pre-project loading: 119.1%

17. Facility: BRUCE CROSSING-HOLOPAW 138 kV Line
Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV
Loading: 124.0% (Rating: 459 MVA)
DFAX: 30.4%
Pre-project loading: 115.4%

18. Facility: PRAIRIE WIND FARM-BELTSVILLE SUBSTATION 138 kV line
Contingency: RIVA RD-SHINGLETOWN 345 kV Line
Loading: 124.0% (Rating: 338 MVA)
DFAX: 11.2%
Pre-project loading: 119.7%

19. Facility: HOLOPAW-PORT ROBSON SUBSTATION 138 kV
Contingency: RIVA RD-SIERRA PACIFIC LUMBER SUBSTATION 345 kV
Loading: 122.9% (Rating: 387 MVA)
DFAX: 17.5%
Pre-project loading: 117.0%

20. Facility: ST CLOUD-COSMIC 138 kV
Contingency: LAKE CREEK-SIERRA PACIFIC LUMBER 138 kV
Loading: 118.5% (Rating: 182 MVA)
DFAX: 9.1%
Pre-project loading: 112.0%

21. Facility: RIVA RD 138/1 kV TX
Contingency: SIERRA PACIFIC LUMBER 138-1 kV Auto
Loading: 114.2% (Rating: 536 MVA)
DFAX: 14.2%
Pre-project loading: 110.8%

22. Facility: PAJUI TC-PORT ROBSON 138 kV
Contingency: RIVA ROAD-SIERRA PACIFIC LUMBER 345 kV
Loading: 112.5% (Rating: 420 MVA)
DFAX: 21.6%
Pre-project loading: 105.8%

23. Facility: GLEN BURNIE-ALLIANCE 138 kV
Contingency: RIVA RD SUBSTATION-SIERRA PACIFIC LUMBER 345 kV Line
Loading: 111.1% (Rating: 263 MVA)
DFAX: 20.6%
Pre-project loading: 100.9%

24. Facility: TERRY-FRENCH GULCH 138 kV
Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV
Loading: 110.5% (Rating: 225 MVA)
DFAX: 7.7%
Pre-project loading: 106.1%

25. Facility: ACEC GRANT-WAUGH CHAPEL 138 kV
Contingency: SIERRA PACIFIC LUMBER-RIVA RD SUBSTATION 345 kV
Loading: 108.0% (Rating: 218 MVA)
DFAX: 13.4%
Pre-project loading: 100.0%

26. Facility: COLUMBIA-SAN LORENZO 3302 138 kV
Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV
Loading: 107.9% (Rating: 218 MVA)

DFAX: 8.7%

Pre-project loading: 102.7%

27. Facility: RALPH SUBSTATION-FENCH GULCH 138 kV

Contingency: SIERRA PACIFIC LUMBER-RIVA RD 345 kV Line

Loading: 106.5% (Rating: 219 MVA)

DFAX: 7.7%

Pre-project loading: 102.0%

28. Facility: SAN LORENZO 3302-PAJUIL TC 138 kV

Contingency: RIVA ROAD-SIERRA PACIFIC LUMBER 345 kV

Loading: 104.6% (Rating: 299 MVA)

DFAX: 13.9%

Pre-project loading: 98.5%

29. Facility: HOOKDALE-TERRY 138 kV L

Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV

Loading: 102.6% (Rating: 285 MVA)

DFAX: 7.7%

Pre-project loading: 99.1%

30. Facility: HOOKDALE-DIXLETA 138 kV Circuit 1

Contingency: RIVA RD-SIERRA PACIFIC LUMBER 345 kV

Loading: 102.4% (Rating: 341 MVA)

DFAX: 7.1%

Pre-project loading: 99.7%

31. Facility: DUNLAVY SUBSTATION-VIADUCT AVENUE 138 kV

Contingency: PORT ROBSON-HOLOPW 138 kV

Loading: 100.1% (Rating: 206 MVA)

DFAX: 6.7%

Pre-project loading: 95.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SANDARIO-FOX RIV 138 kV line	Rebuild line with higher-capacity conductor	\$5,178,000	30% (\$1,554,000)
SIERRA PACIFIC LUMBER-CALIFORNIA AVE 138 kV L	Rebuild line with higher-capacity conductor	\$12,949,000	30% (\$3,886,000)
CHANNELVIEW-CALIFORNIA AVE SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,305,000	30% (\$692,000)
CHANNELVIEW SUBSTATION-GEORGIA-PACIFIC CROSSETT LLC 138 kV	Rebuild line with higher-capacity conductor	\$2,308,000	30% (\$693,000)
FOX RIV-GEORGIA-PACIFIC CROSSETT LLC 138 kV L	Rebuild line with higher-capacity conductor	\$2,619,000	30% (\$786,000)
FOOTHILL-WAUGH CHAPEL 138 kV	Rebuild line with higher-capacity conductor	\$5,231,000	34% (\$1,776,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GLEN BURNIE-SANDARIO 138 kV	Rebuild line with higher-capacity conductor	\$2,756,000	18% (\$490,000)
FOOTHILL-MCCALL SUBSTAION 138 kV	Rebuild line with higher-capacity conductor	\$1,821,000	34% (\$618,000)
HUNT CLUB-SIERRA PACIFIC LUMBER 138 kV line	Rebuild line with higher-capacity conductor	\$12,611,000	41% (\$5,126,000)
HUNT CLUB-SALAMANCAFRANK ST 138 kV Line	Rebuild line with higher-capacity conductor	\$2,365,000	41% (\$961,000)
STILLWATER HYDRO ELECTRIC PROJECT-SANDARIO 138 kV	Rebuild line with higher-capacity conductor	\$1,187,000	39% (\$467,000)
MCCALL SUBSTAION-PEBBLEDALE SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,322,000	37% (\$863,000)
ALLIANCE SUBSTATION-RO BLANCO 2702 SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$1,733,000	34% (\$580,000)
MCCALL SUBSTAION-STILLWATER HYDRO ELECTRIC PROJECT SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,523,000	39% (\$993,000)
BRUCE CROSSING SUBSTATION-RIO BLANCO 2702 138 kV	Rebuild line with higher-capacity conductor	\$2,782,000	34% (\$931,000)
TABER-PEBBLEDALE 138 kV	Rebuild line with higher-capacity conductor	\$2,774,000	37% (\$1,030,000)
BRUCE CROSSING-HOLOPAW 138 kV Line	Rebuild line with higher-capacity conductor	\$4,503,000	34% (\$1,507,000)
BELTSVILLE-PRAIRIE WIND FARM 138 kV	Rebuild line with higher-capacity conductor	\$6,966,000	14% (\$948,000)
HOLOPAW-PORT ROBSON SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$1,745,000	30% (\$532,000)
ST CLOUD-COSMIC 138 kV	Rebuild line with higher-capacity conductor	\$5,848,000	41% (\$2,377,000)
RIVA ROAD 138/1kV TX	Replace transformer with higher-rated unit	\$6,394,000	8% (\$507,000)
PAJUIL TC-PORT ROBSON 138 kV	Rebuild line with higher-capacity conductor	\$1,880,000	37% (\$695,000)
ALLIANCE-GLEN BURNIE 138 kV	Rebuild line with higher-capacity conductor	\$2,691,000	31% (\$836,000)
TERRY-FRENCH GULCH 138 kV	Rebuild line with higher-capacity conductor	\$3,451,000	41% (\$1,425,000)
ACEC GRANT-WAUGH CHAPEL 138 kV	Reconductor line	\$2,412,000	34% (\$819,000)
COLUMBIA-SAN LORENZO 3302 138 kV L	Reconductor line	\$1,589,000	41% (\$656,000)
FRENCH GULCH SUBSTATION-RALPH 138 kV	Reconductor line	\$659,000	41% (\$272,000)
SAN LORENZO 3302-PAJUIL TC 138 kV	Reconductor line	\$379,000	37% (\$140,000)
TERRY SUBSTATION-HOOKDALE 138 kV Line	Reconductor line	\$674,000	41% (\$278,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HOOKDALE SUBSTATION-DIXLETA 138 kV Ckt 1	Reconductor line	\$605,000	25% (\$151,000)
DUNLAVY-VIADUCT AVENUE 138 kV	Reconductor line	\$624,000	14% (\$90,000)
	Total allocated to project		\$32,679,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.